# STARBOOSTER

# RAGTIME

Project Manager: Spencer Biernat



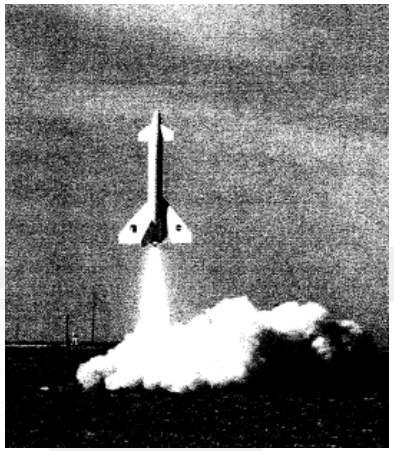
# AGENDA

1. Introduction
2. Mission Requirements
3. Vehicle Overview
4. Concept of Operations
5. Aerodynamic Analysis
6. Subsystem Designs
    1. Lift and Control Surfaces
    2. Structures and Recovery
    3. Avionics and Radio Control
    4. Telemetry and Ground Station
7. Testing
8. Next Steps

# BACKGROUND

## Background on the Booster Project

- CPSS has history with flyback boosters: StarBooster. CPSS booster team's Sonata 2 project aims to improve sonata 1's design and prove stable, replicable flight for implementation with a second booster segment.
- Sonata I is a vertical launch spaceplane conceptualized as a test platform for systems and component designs for eventual inclusion in later Opus program projects.
- Because Sonata I is primarily a test platform, modularity, manufacturability, and ability to quickly relaunch are the main focuses in its design.
- The CPSS recoverable booster team's long-term goal with the Opus program is to develop a flyback solid rocket booster system that can serve as a platform to launch future CPSS projects such as propulsive landers, space shots, or other multi-stage projects. Sonata I is the first step in developing the systems, manufacturing processes, operational skills, and infrastructure to support and maintain such a complex system.



Final report*: https://ntrs.nasa.gov/api/citations/20020023126/downloads/20020023126.pdf





# OVERVIEW

## Mission Scope

- **Mission Objective:** Launch the vehicle as a rocket, land the vehicle as a glider.
- **Mission Success Criteria**

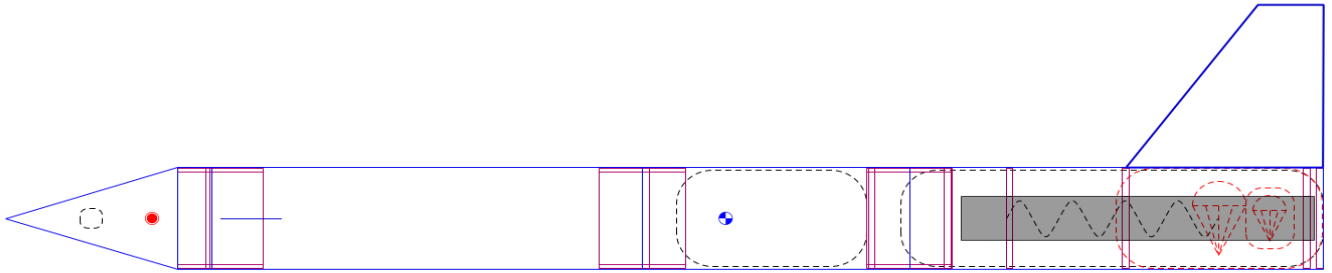
1. The vehicle reaches an apogee of between 1000-2000 feet.
2. The vehicle lands after a successful glide phase
    - Lands in a way that the vehicle could hypothetically be reused.

# OVERVIEW

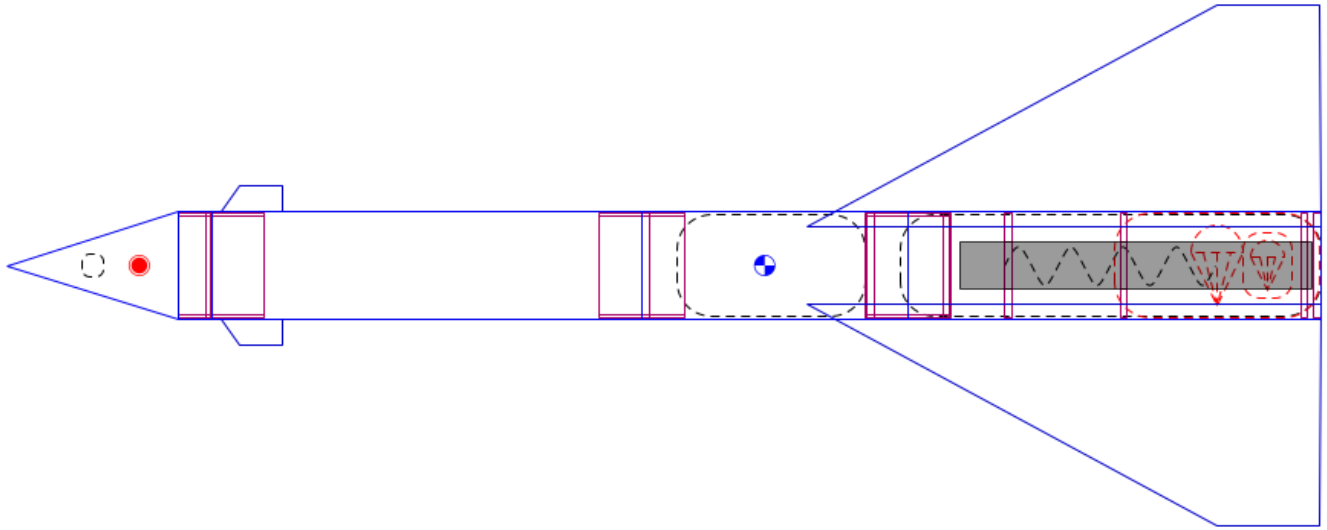
Open Rocket + mass/dimension estimates

- **Estimated Loaded Mass:** 26.55 lbs*
- **Estimated Unloaded Mass:** 23.12 lbs
- **Overall Length:** 56.70”
- **CP Location:** 15.75” from rear
- **Stability:** Unknown

Side view:

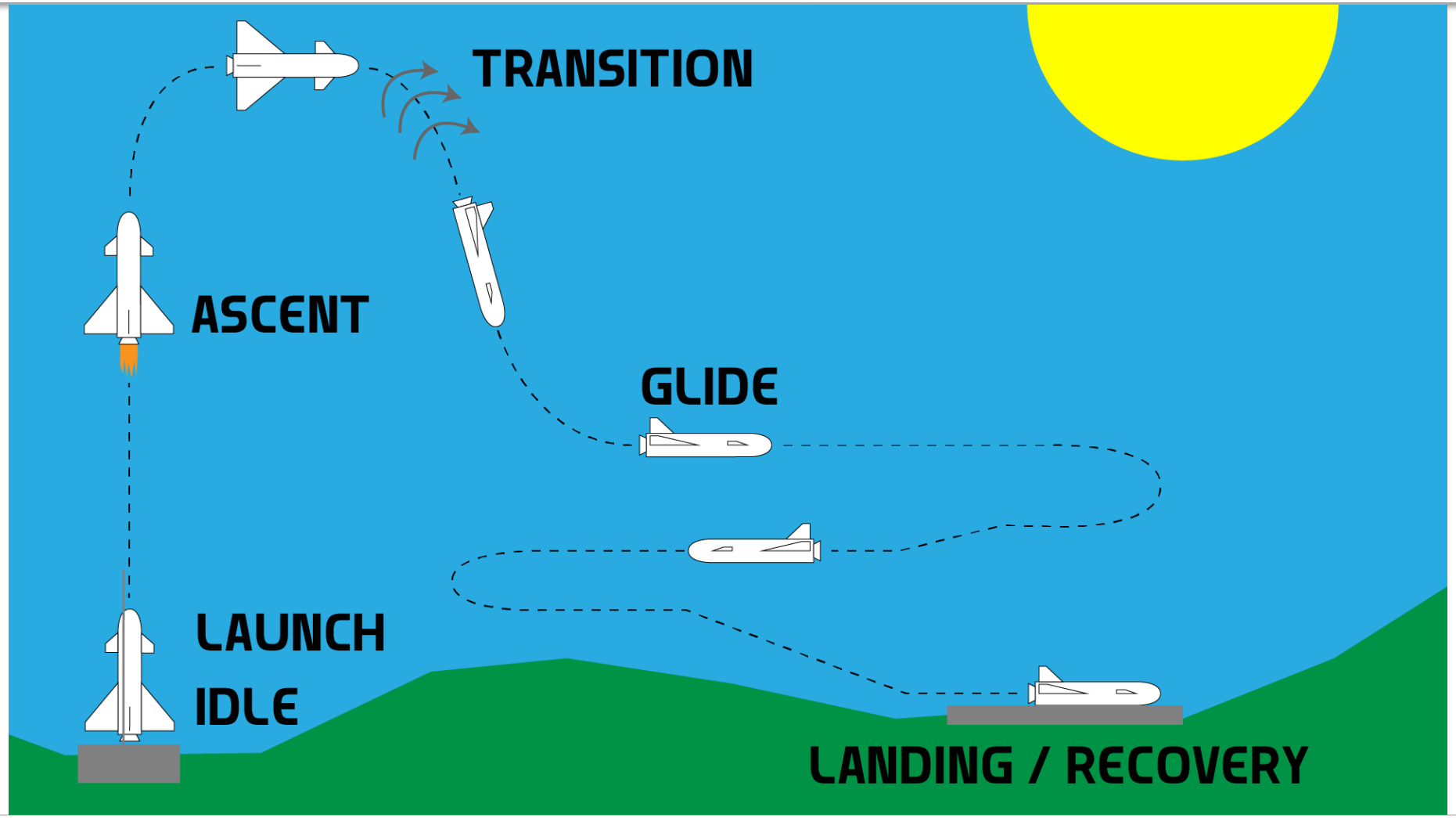


Top view:



# CONOPS

Graphic



# CONOPS PHASES

Phases

- Idle
  - Begin transmitting telemetry, video feed, and data logging. Video feed is being transmitted to remote operator.
- Launch
  - Motor ignition.
- Ascent
  - Flight software will be configured to maintain stable upward trajectory with automatic adjustments of control surfaces. Adjustments will only be minor as vehicle should be naturally stable.
  - The operator will be on standby and only observe.
- Transition
  - Authority over control surfaces is handed off to remote operator. Also given the option to abort from now on.
  - Water payload released.
- Glide
  - Operator glides the vehicle towards landing site, using the video feed to see and controller to operate.
  - Glide slope will be very steep, interrupted by periods of leveling off to bleed off speed.
  - Flight software will assist with automatic stability corrections when needed.
- Landing / Recovery
  - Operator safely lands the vehicle at the landing site.

# REQUIREMENTS

Top Level

| RBS | Requirement | Rationale |
|---|---|---|
| 1 | The vehicle shall reach an apogee of between 1000-2000 feet. | This apogee is needed for the vehicle to have the altitude necessary to glide and land. Also, a success criteria. |
| 2 | The vehicle shall have fixed wings. | Wings are needed for the vehicle to glide. |
| 3 | The vehicle shall make a controlled descent and landing. | A controlled descent is necessary for the glide phase to be successful. |
| 4 | The vehicle shall remain structurally stable during flight and landing. | Mission failure will occur if vehicle breaks apart during those phases. |
| 5 | The vehicle shall remain aerodynamically stable during all flight phases. | Abort conditions would be met if vehicle becomes uncontrollable. |
| 6 | The vehicle shall transmit telemetry during the flight. | To collect and relay data/video to the ground operator so they can operate the vehicle. |
| 7 | The vehicle shall be legal under FAA/Local regulations and ordinances | The vehicle is to be launched at a site in the local area—not the desert, so adhering to FAA/Local regulations is important. |

# AERODYNAMIC ANALYSIS

Spencer Biernat, Shiva Pulmamidi, David Carter Oropeza

# DELTA WINGS

Purpose:

- Delta wings were chosen in place of regular plane wings, due to the former minimizing drag until apogee. Additionally, it provides more structural stability during takeoff, when acceleration and force on the wings is greatest. The delta wings will be the main generators of lift which they will perform by servo-controlled ailerons

Lift= (1/3)1036.84 lbf* = Cl x S x (½) x p x v^2

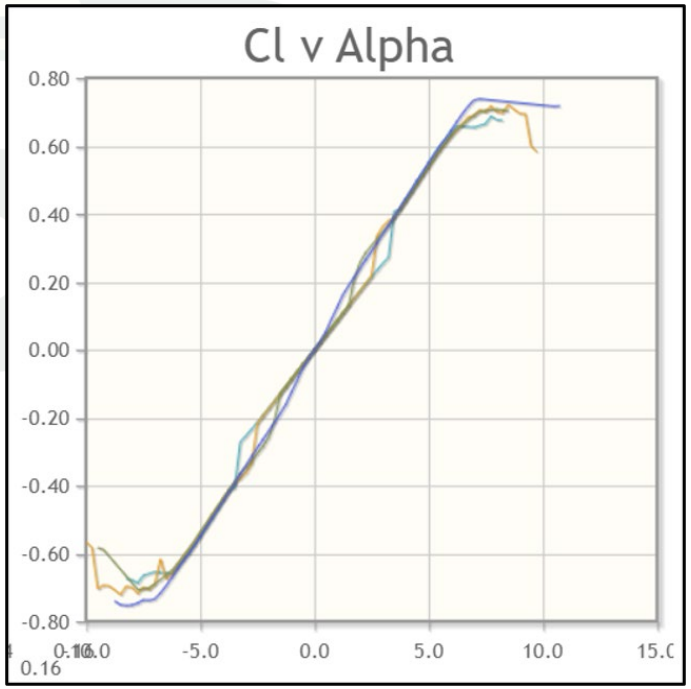
-Cl= Coefficient of Lift= 0.55

-S= Wing surface Area*= 18.254 ft^2

-P= density (sea level)= 0.0765 lb/ft^3

-V= 30 ft/s

*To be determined/tentative

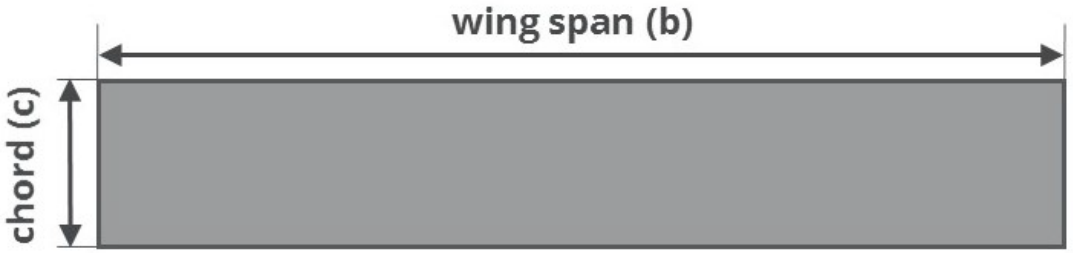



NACA 0006 Airfoil

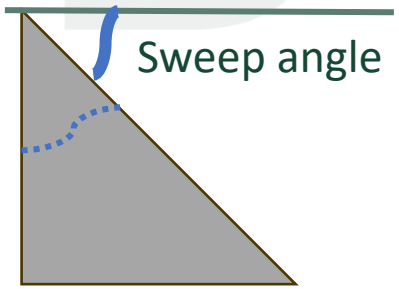
# DELTA WINGS

## Wingspan Calculation

- Aspect Ratio (A): tan(90-40) ~ 1.19
- Wing Area (S)*: 18.254 ft^2
- Wingspan*= sqrt(A x S) ~ 4.66 ft
- Base of each wing* ~ 2.33 ft
- Root Chord* ~ 1.96 ft




**Aspect Ratio:** $$AR = \frac{b}{c} = \frac{b}{c} \times \frac{b}{b} = \frac{b^2}{A}$$




c= 1.96 ft

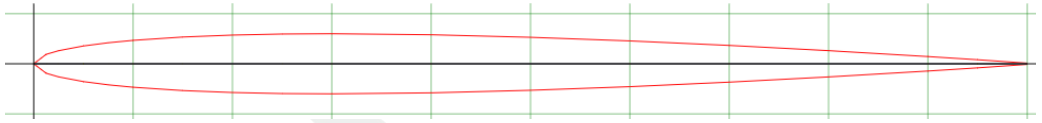
b= 2.33

*To be determined/tentative

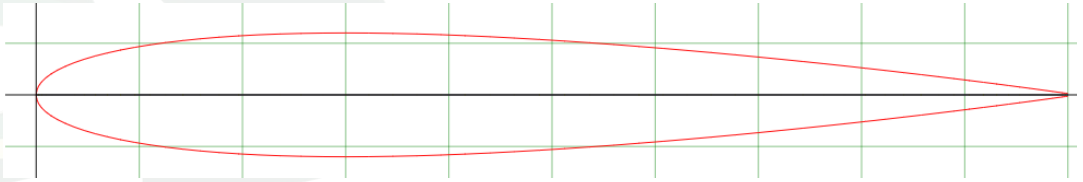
# VERTICAL STABILIZER AND RUDDER

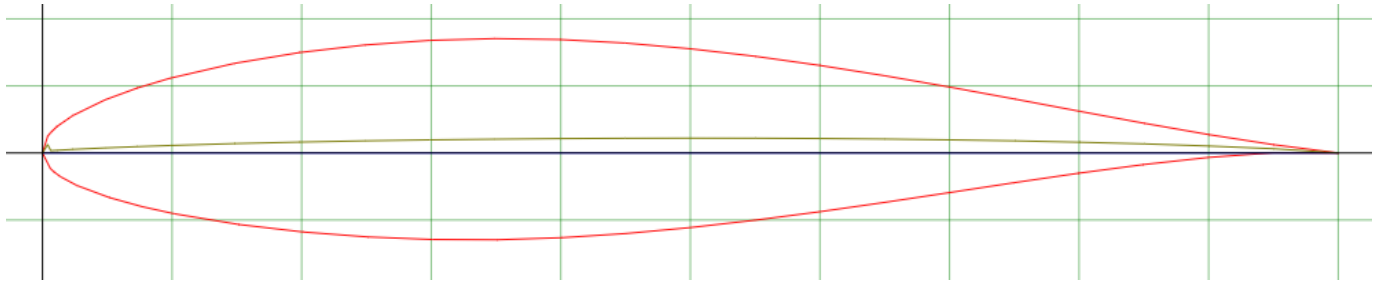
- Purpose of Vertical Stabilizer and Rudder.
- Main problem with Sonata I.
- CAD is currently using NACA 63-215 AIRFOIL, this design will change.
- We will most likely use a NACA 0006 or 0012 design in the future.

NACA 0006:



NACA 0012:





# SERVOS

- The servos to control the ailerons were chosen due to two main parameters: The clearance allowed inside the rocket, and the estimated required torque based on the expected velocity and control surface sizes given by the following equation:

$$Torque\ (oz.-in) = 8.5 \times 10^{-6}\left(C^2 \times V^2 \times L \times \sin(S1) \times \frac{\tan(S1)}{\tan(S2)}\right)$$

- C = control surface chord (cm)
- L = control surface length (cm)
- V = velocity (mph)
- S1 = max control deflection in degrees
- S2 = max servo deflection in degrees



# CANARDS & VERTICAL STABILIZER

Canards:

- Since the wing is so far back on the vehicle's body, canards are used to provide pitch stabilization and elevator control on the way down. For their design they are using the same airfoil and shape as the wing but sized down to 20% of the wings size (as is generally standard). The old StarBooster had each canard sized at about 15% of the wings, so the design is kept in similarity with this. The canards on the old booster were angled upwards slightly, which would create some upward lift on the vehicle. To simplify the aerodynamics, the current design will not feature this. However, the canards will have elevators that will be actuated by an in-line servo directly connected to the elevator.

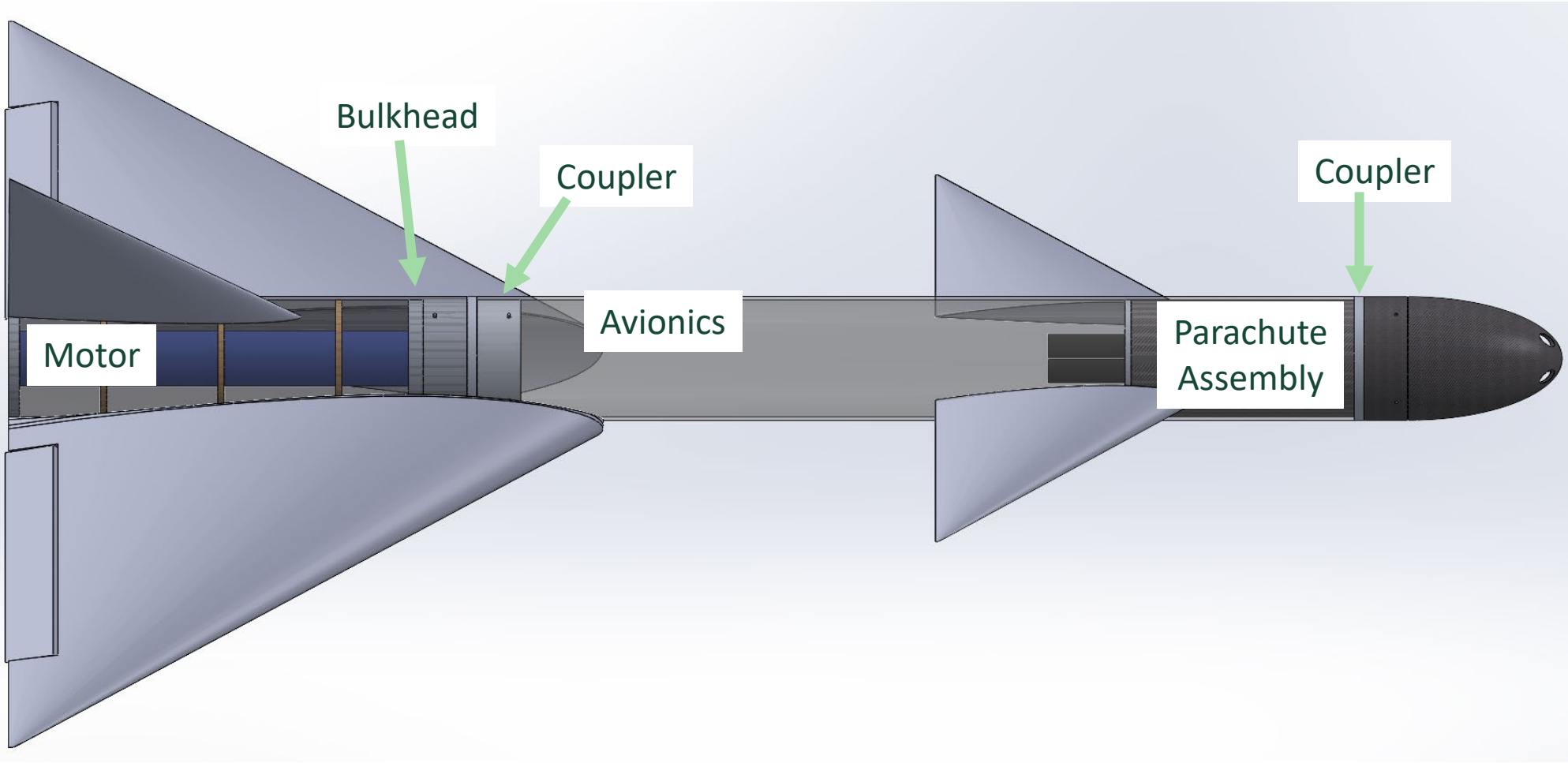
Vertical Stabilizer:

- The vertical stabilizer on the old booster was about 43% the size of one wing. This is pretty big and was likely designed to offset the delta wings mounted on the bottom of the rocket's body. Since rudder control is not a big concern as compared to weight, the vertical stabilizer is currently sized at 20% of the wing. As the design is meant to be modular then this can be changed out if yaw control/stability is found to be too poor.

# STRUCTURES

Jared Lebaron, David Carter Oropeza, Shiva Pulmamidi, Alistair Hynes

# OVERVIEW


Bulkhead
Coupler
Coupler
Avionics
Motor
Parachute
Assembly

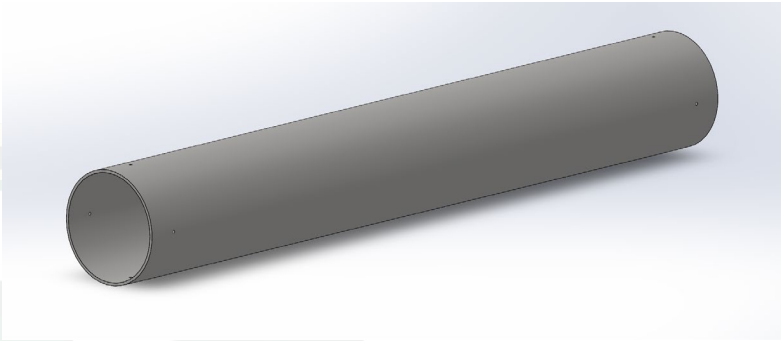
# BODY TUBES

## LOWER BODY TUBE



| | |
|---|---|
| Material: | Carbon Fiber |
| Length: | 18.51” |
| Outer Diameter: | 5” |
| Inner Diameter: | 4.75” |
| Thickness: | .125” |
| Estimated Weight: | 2.56 lbs |

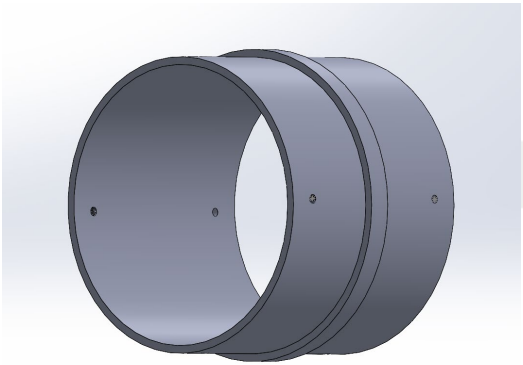
## UPPER BODY TUBE



| | |
|---|---|
| Material: | Fiberglass |
| Length: | 29.40” |
| Outer Diameter: | 5” |
| Inner Diameter: | 4.75” |
| Thickness: | .125” |
| Estimated Weight: | 5.63 lbs |

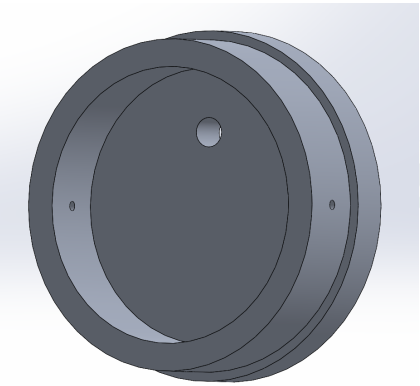
# INNER COMPONENTS

## COUPLERS (x2)



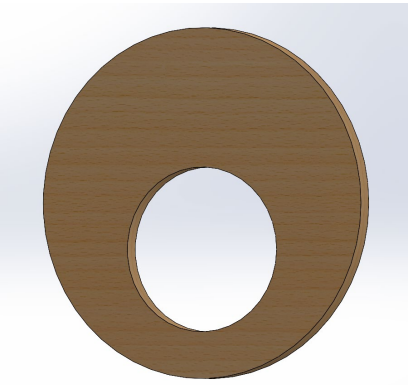
| | |
|---|---|
| Material: | Aluminum |
| Length: | 3.94” |
| Ridge Diameter: | 5” |
| Outer Diameter: | 4.75” |
| Inner Diameter: | 4.5” |
| Thickness: | .125” |
| Weight: | .770 lbs |

## BULKHEAD



| | |
|---|---|
| Material: | Aluminum |
| Length: | 1.597” |
| Outer Diameter: | 4.75” |
| Inset Diameter: | 4.5” |
| Inner Diameter: | 3.75” |
| Hole Diameter: | .40” |
| Hole Offset: | 1.00” |
| Weight: | 1.50 lbs |

## CENTERING RINGS (x2)



| | |
|---|---|
| Material: | Wood |
| Length: | .25” |
| Outer Diameter: | 4.75” |
| Hole Diameter: | 2.23” |
| Hole Offset: | -.625” |
| Weight: | .02 lbs |

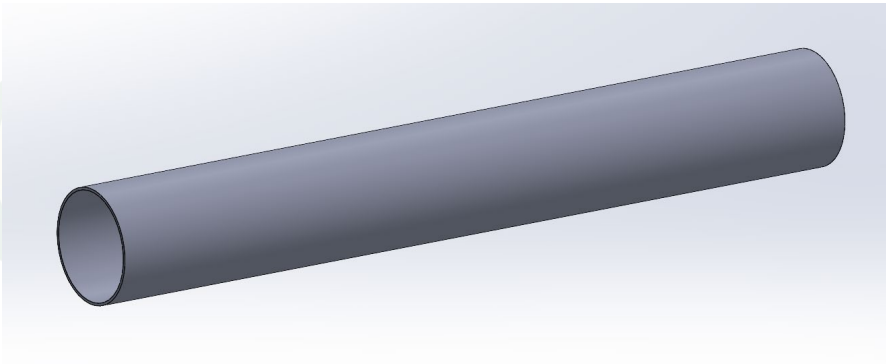
# MOTOR-RELATED COMPONENTS

## MOTOR RETENTION PLATE



| | |
|---|---|
| Material: | Aluminum |
| Length: | .45” |
| Outer Diameter: | 4.75” |
| Hole Diameter: | 2.23” |
| Hole Offset: | -.625” |
| Weight: | .51 lbs |

## MOTOR TUBE



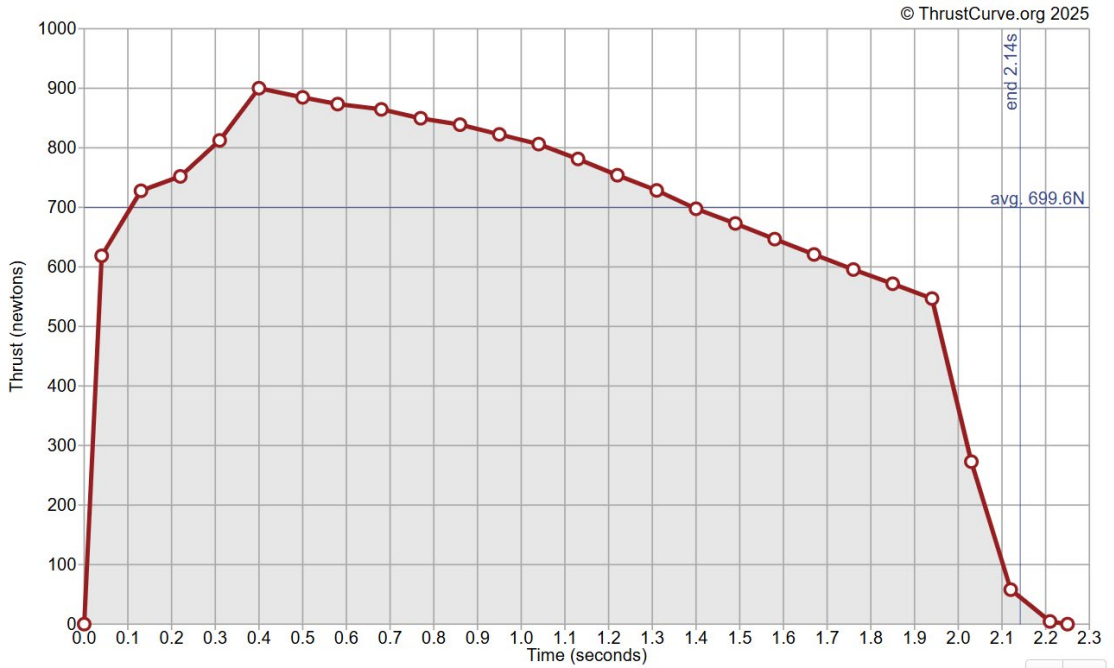
| | |
|---|---|
| Material: | TBD |
| Length: | 16.142” |
| Outer Diameter: | 2.23” |
| Inner Diameter: | 2.126” |
| Thickness: | 0.052” |
| Weight: | .56 lbs |

# MOTOR

Aerotech K695 Motor (Model 81445)

- Diameter: 2.126”
- Length: 16.142”
- Total Weight: 3.278 lbs
- Propellant Weight: 1.991 lbs

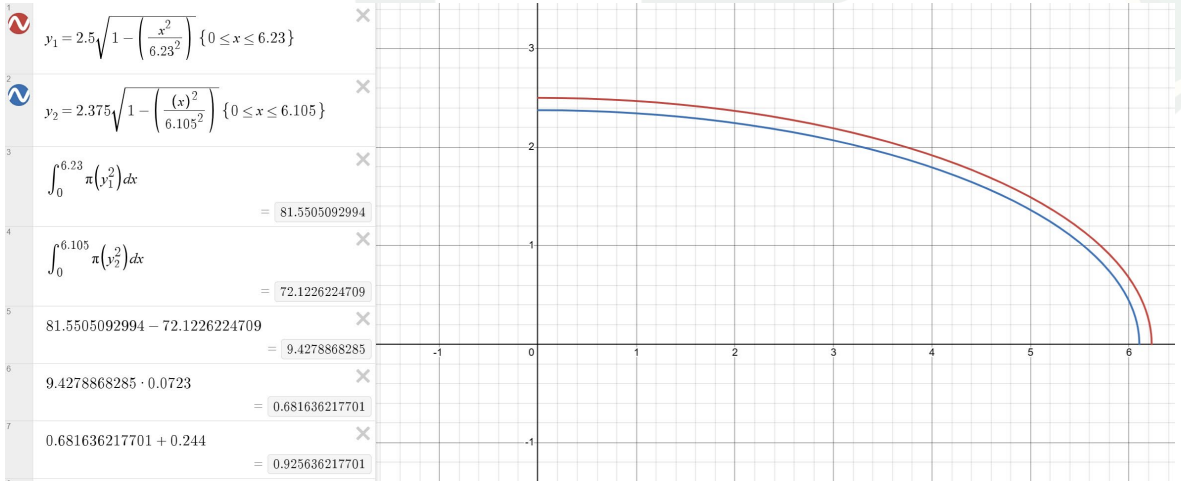
- Average Thrust: 695 N
- Initial Thrust: 759.7 N
- Total Impulse: 1514 Ns
- Burn Time: 2.2 s
- Delay: 14 s

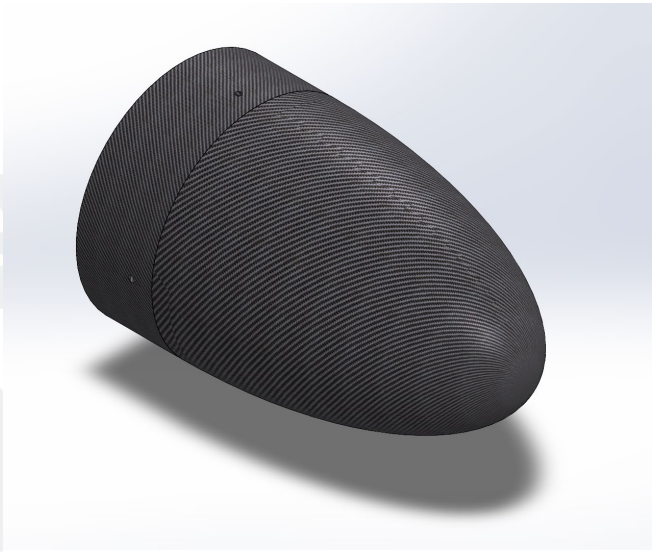



- Chosen for relatively consistent thrust curve, reducing maximum acceleration and impulse on the wings
- Aerotech K695 allows us to reach our target apogee and is available for purchase

# NOSE CONE

- Based on elliptical formula $Y = R\sqrt{1 - \left(\frac{x^2}{L^2}\right)}$, which is optimal for subsonic speeds and provides ample room for the integration of a camera system
- Water payload will be placed in nose cone, released after apogee
- Weight calculation shown below (SolidWorks weight from Mass Properties was inaccurate)



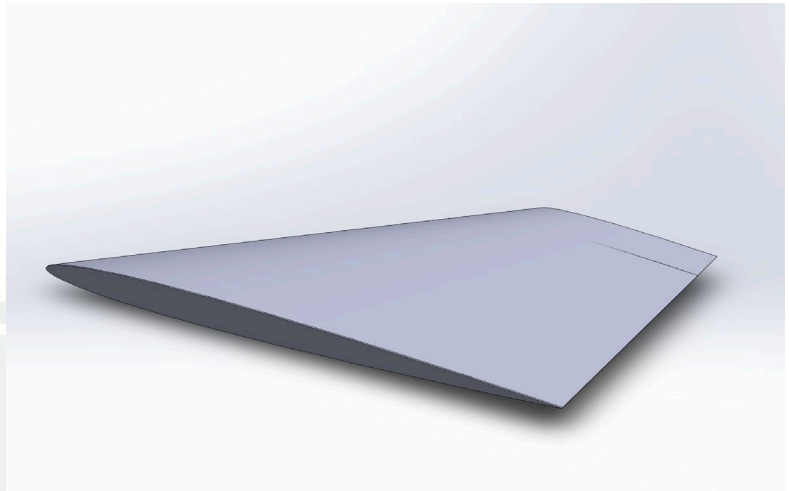



| | |
|---|---|
| Material: | Carbon Fiber |
| Length: | 8” |
| Outer Diameter: | 5” |
| Inner Diameter: | 4.75” |
| Weight: | 0.926 lbs |

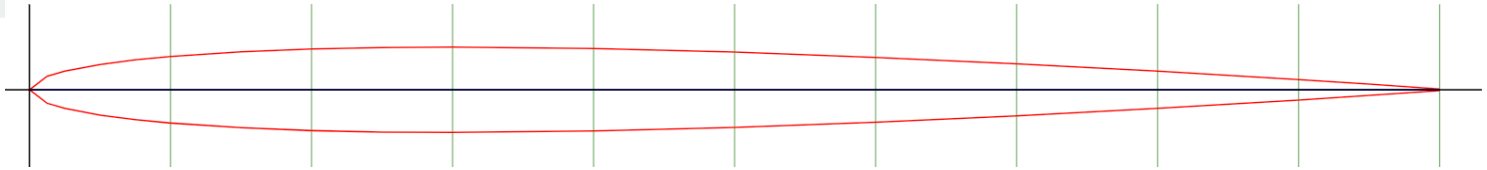
# WINGS (DELTA)

NACA 0006 Airfoil- Chosen for its low thickness to minimize drag

- Material: TBD
- Root Chord ~1.96 ft*
- Tip Chord: TBD
- Wing Area ~9.127 ft^2
- Wing Aspect: 1.19
- Wing Loading: 1.46 lb^2/ft^2
- Base of wing ~ 2.33*



*Not made to dimensions on the left

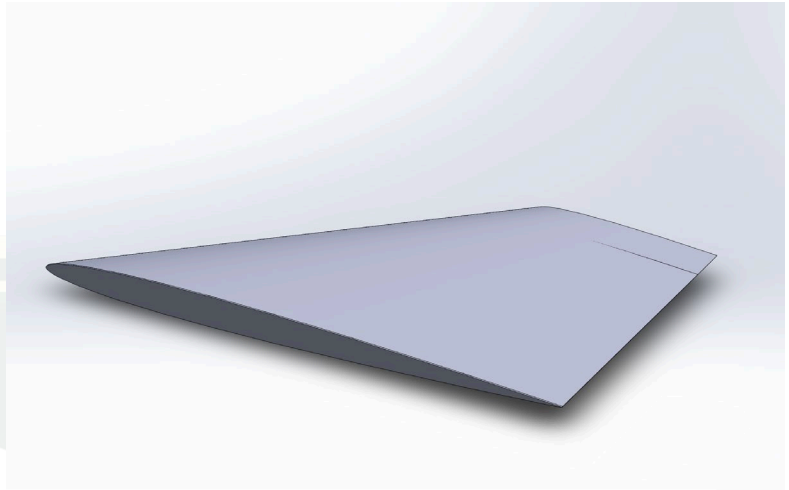


*Subject to change/Have already changed

# WINGS (CANARDS)

NACA 0006 Airfoil

- Material: TBD
- Root Chord ~6.89 in*
- Tip Chord ~2.36 in*
- Wing Area ~1.94 in^2*
- Wing Aspect: 2.6*
- Wing Loading: 1.26*
- Wingspan ~4.722 in*



*Subject to change

# WING "MOUNT"

The current wing mounting design follows the approach used in Sonata 1. While this provides a starting point, we recognize potential shortcomings due to increased aerodynamic and landing loads for the glider configuration. As development progresses, we will evaluate more robust alternatives. Potential alternatives we have investigated are listed below.

- **Through-the-Wall (TTW) Attachment:** Integrating structural tabs from the wing root through pre-cut slots in the fuselage, bonding to internal structural supports.
- **Internal Spar Pass-Through:** Extending the wing's internal spar into the rocket body to directly transfer aerodynamic and landing forces into the airframe.
- **Reinforced Hardpoint Mounting:** Embedding load-bearing plates (e.g., carbon fiber, G10, or aluminum) into the wing root and fuselage to distribute attachment forces more evenly.
- **One Piece wing with rocket body slot:** A single delta wing will be manufactured with a central slot for the rocket fuselage. The wing will have a foam core, fiberglass skin, internal spars, and reinforced slot edges. It will be bonded and/or bolted to the fuselage for strength and load distribution.

# SERVO PLACEMENT

Servo placement is being finalized with a focus on simplicity and proper weight distribution. For the canards, we plan to place the servos inside the rocket body, adjacent to the canard pivot point, allowing for a protected and direct actuation system with minimal linkage complexity. The canard will function as a full-span, actuated control surface.

For the main wings, servos will be mounted flush within the wing structure to reduce control rod distance and simplify mechanics. During manufacturing, a slot will be cut into the wing along the hinge line between the wing and the control surface for servo installation, with the control surface temporarily removed. This approach provides a standard, reliable configuration with good aerodynamic performance.

We will continue evaluating these placements to ensure they meet performance, reliability, and manufacturability requirements.

# VERTICAL STABILIZER AND RUDDER

Material: TBD

Airfoil Design: NACA 63-215 AIRFOIL*

Thickness: 1.5"* (Root Chord)

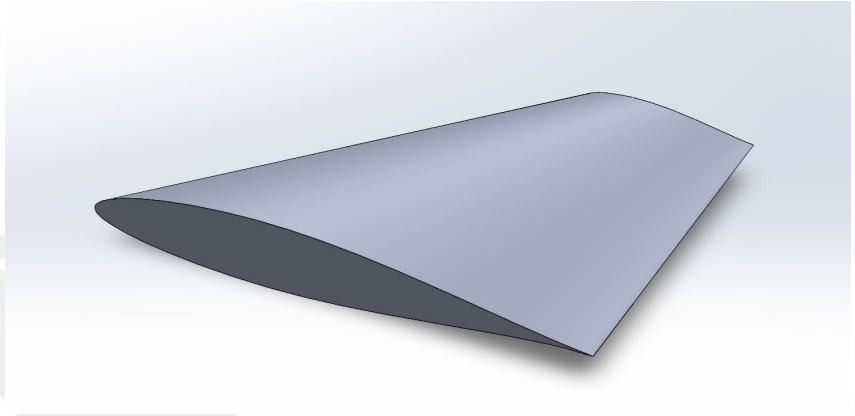
Tip Chord: 4”*

Root Chord: 12”*

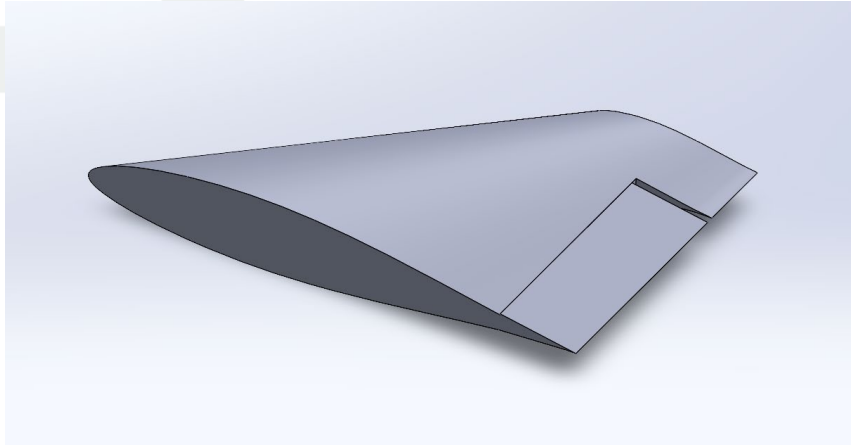
Height: 8"*

*Subject to Change

Vertical Stabilizer:

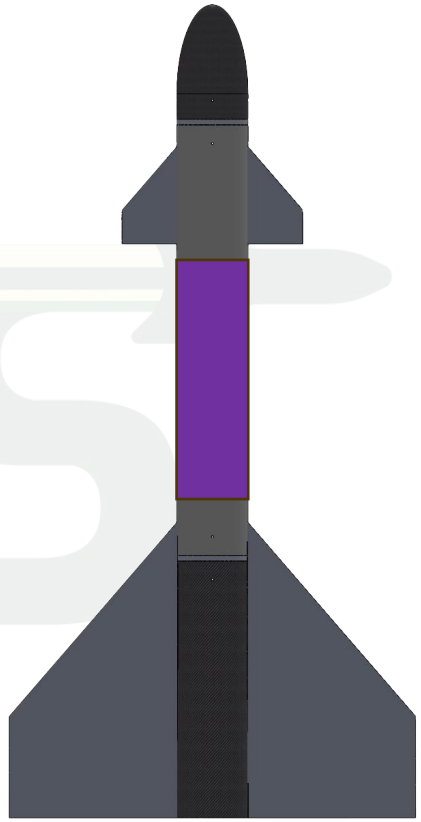


Vertical Stabilizer with Rudder:



# SKID PLATE

- To prevent damage to the body when landing adding a skid plate to the belly of the rocket is necessary.
- The skid plate would have to be easily removable in respect to the surface the rocket is landing on.
- The skid plate would be replaced after every launch if need be.



Bottom view of rocket tube

# MATERIALS OVERVIEW

- Carbon Fiber
  - Very strong and stiff, but also lightweight. Helps with trying to make weight with the payload.
- Fiberglass
  - Allows radio waves to pass through for communication
- 6061 Aluminum
  - Affordable and easy-to-machine stock

# RECOVERY

David Carter Oropeza

# PARACHUTE

- The parachute is a safety precaution in case anything goes wrong.
- It will be stored in the back of the rocket next to the motor.
- Basic 60" toroidal parachute weighing 6.8oz, packed 3.9in (D) by 3.2in (L), able to slow 19lb rocket to 20fps. Not the finalized choice for parachute, especially considering it costs $266 and the diameter of its container is too large.

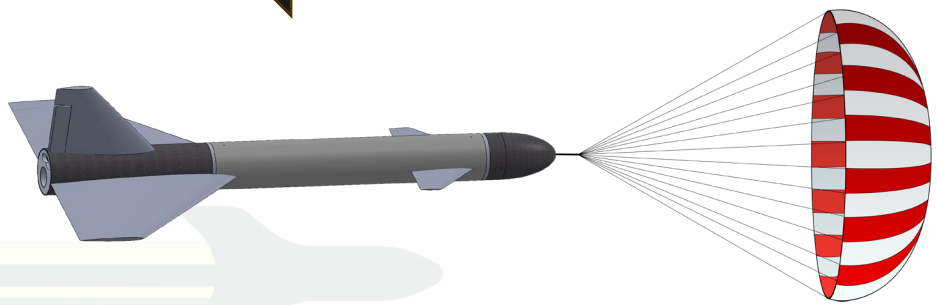




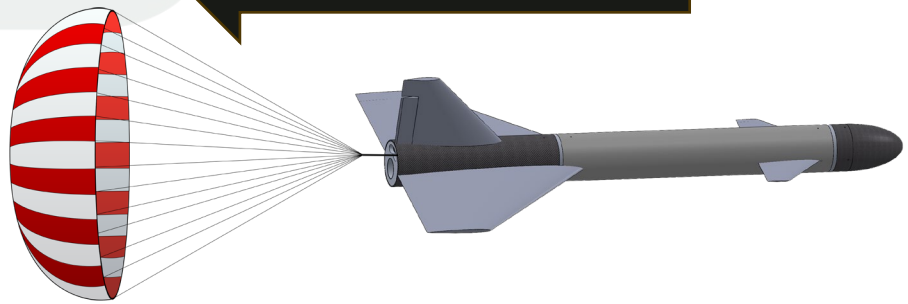

# WHEN TO USE PARACHUTE

- Parachute deployed out the back during free fall.
- Parachute deployed out the back while gliding.
- Activation and reusability of the parachute.
- Instances to use parachute:
  - Wing servos stop working and we lose control of the rocket.
  - Connection to rocket is lost for extended period of time.
  - We are not able to get the rocket to glide at a reasonably safe speed for landing.

Air against the parachute, not good



Air going into the parachute, good



# AVIONICS

Spencer Biernat

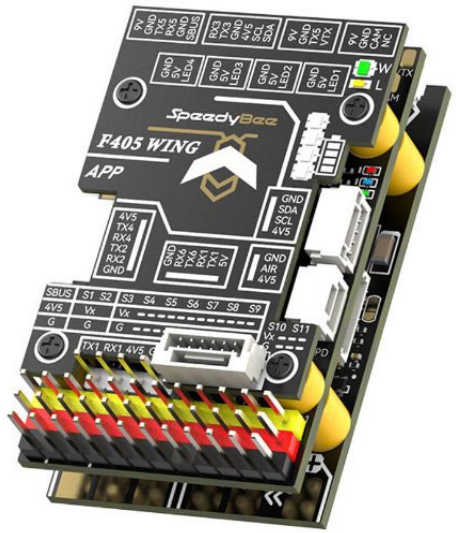
# TELEMETRY AND GROUND STATION: RADIO TRANSMITTER

- RadioMaster TX12 Multi-Protocol OpenTX 2.4GHz RC Transmitter
- Selected because it comes with OpenTX software preinstalled
- Also selected for cost.

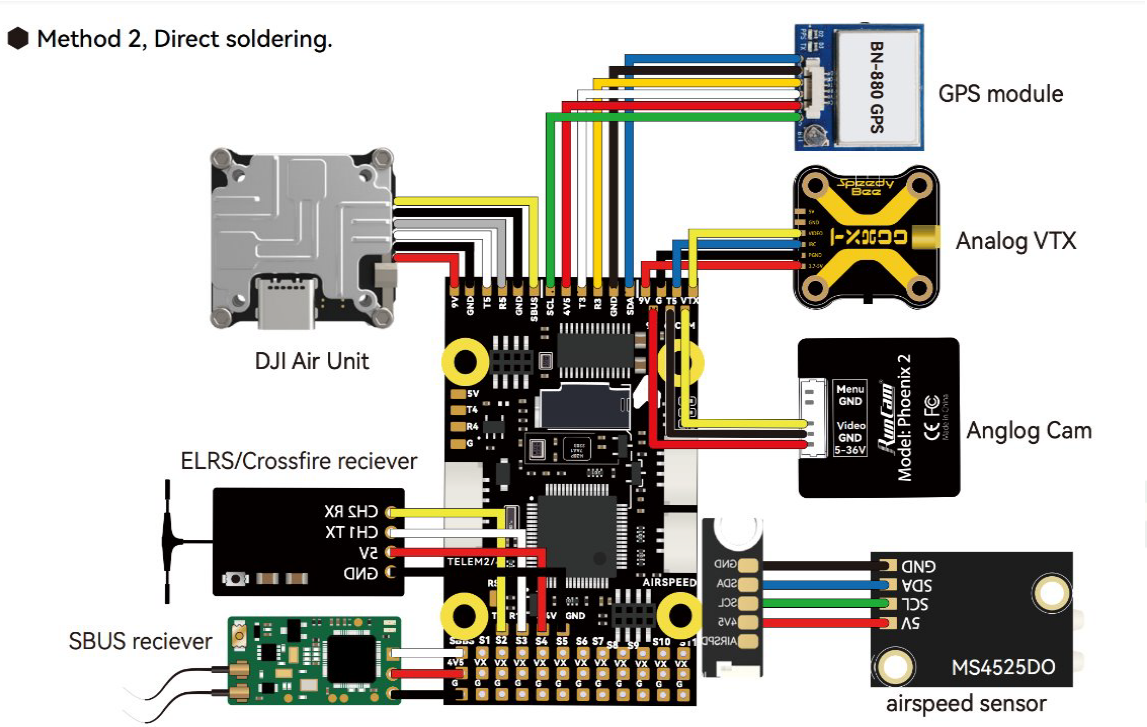


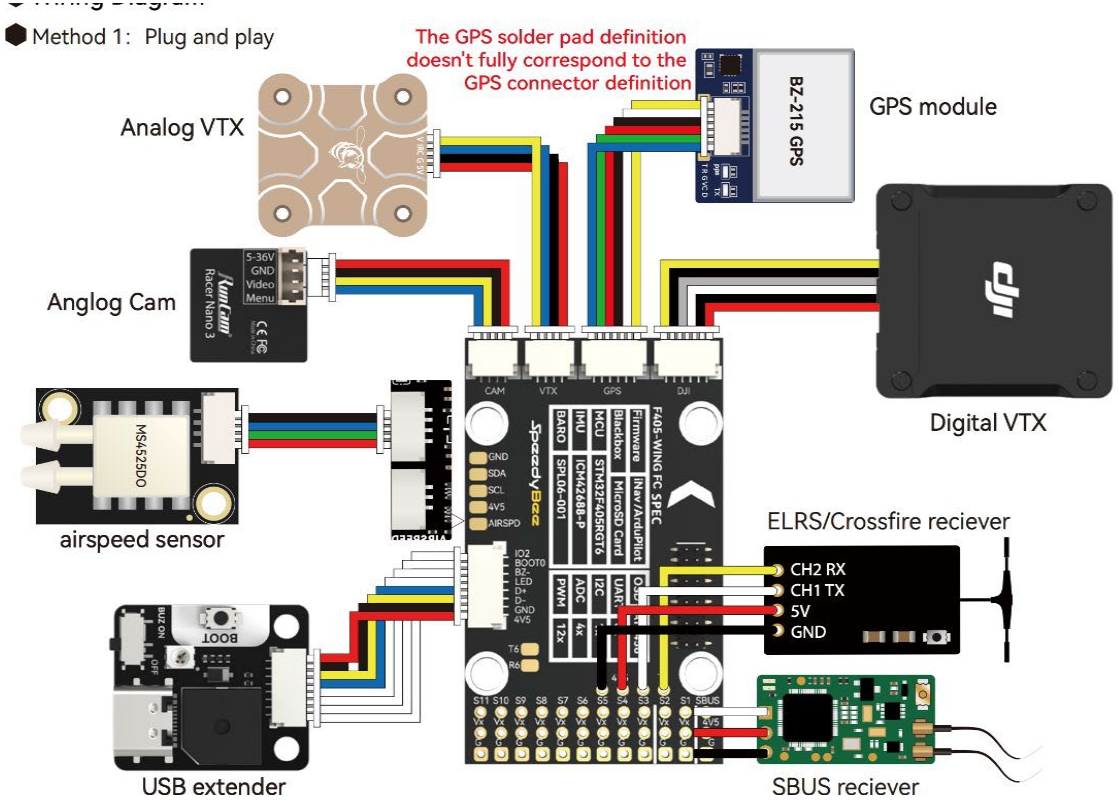
# AVIONICS AND RADIO CONTROL: FLIGHT CONTROLLER

- SpeedyBee F405 WING APP Fixed Wing Flight Controller
- Picked for size, price, and features
- Comes with power distribution board in stack
- dedicated pinouts for all Sonata I components
  - Airspeed sensor
  - GPS
  - Camera and Video TX
  - EdgeTX/OpenTX ready port for telemetry
- Firmware profiles for fixed-wing flight with F405 WING already exist in ArduPilot.
- Plenty of servo control pinouts
- Blackbox logging micro SD card slot.



# AVIONICS AND RADIO CONTROL: WIRING DIAGRAMS



# AVIONICS AND RADIO CONTROL: FLIGHT SOFTWARE

- ArduPilot selected for versatility and availability for integration with future CPSS projects.
- Integration with ground control displays developed using OpenMCT.
- Mission Planner for future projects involving autonomous return.






# TESTING

Spencer Biernat

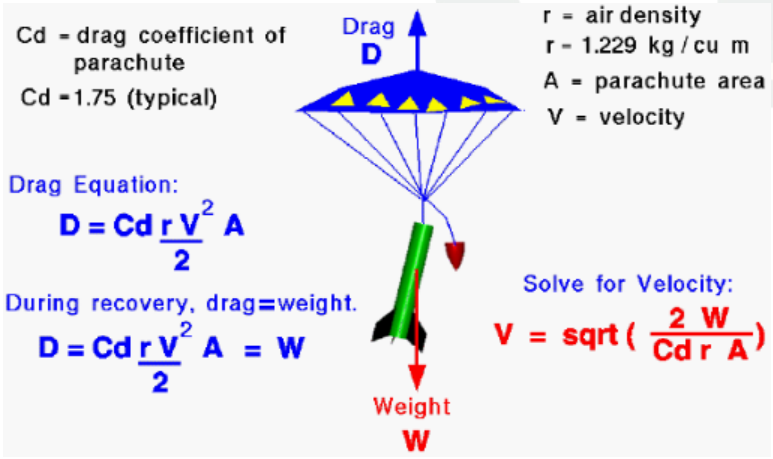
# WIND TUNNEL TEST MATRIX

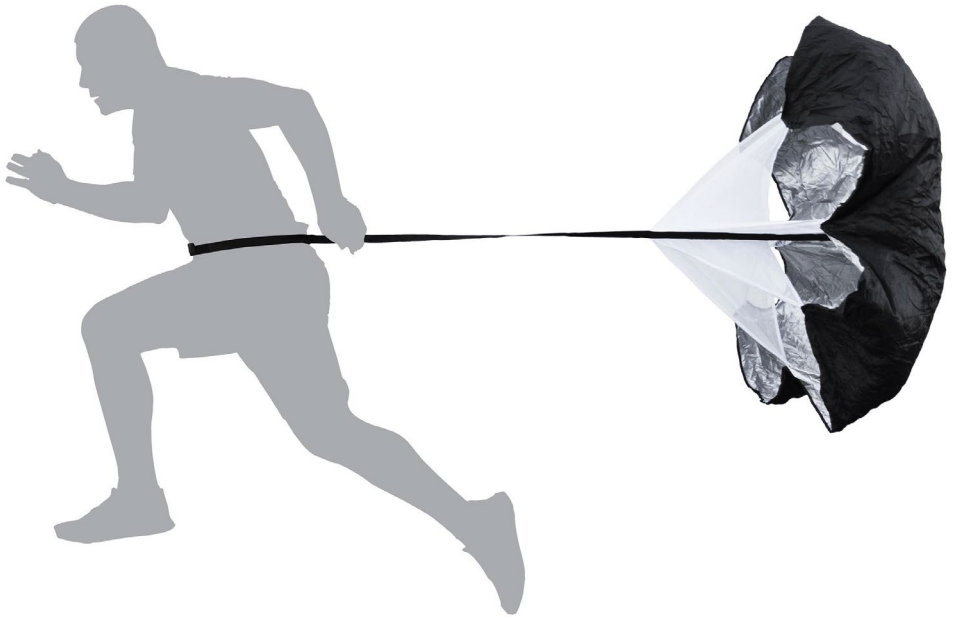
wind

| Test | Description | Data Recordings | Rationale |
|---|---|---|---|
| WT01 | 50% to 100% speed with NACA s7055 delta wings mounted and canards | -Overall drag<br>-airflow over the main wings & canards.<br>-Lift created with 50% -100% elevator deflection. | -Will test how the canards may interfere with the main wing's airflow.<br>-Can fit a curve to the overall drag of the model.<br>-Can fit a lift curve to the wing created with elevator deflection |
| WT02 | Repeat WT01 but with the thicker canard models | -Overall drag<br>-airflow over the main wings & canards.<br>-Lift created with 50% -100% elevator deflection. | Will test how the thicker canards may interfere with the main wing's airflow. |
| WT03 | Repeat WT01 but with a different (thicker) canard model | -Check for overall drag<br>-Airflow over the delta wings and canards<br>-Lift created with 50%-100% elevator deflection | The canards and the delta wings are the main control surfaces and testing is required to see how they will perform in various conditions. |

# PARACHUTE TESTING

- Testing both types of parachute deployment.
- Finding the strength of parachute (drag coefficient).
- We will test the strength at different wind speeds.



# SCALING UP FROM SONATA



- Since Ragtime is essentially a scaled-up version of Sonata, we can take data and lessons learned from Sonata's testing and implement them into Ragtime, much like the scale models tested during the design of the original StarBooster.
- Since Sonata still hasn't been launched, we don't know what needs to be improved on yet. However, we can use the original StarBooster's testing to give us an idea. Possible alterations may include:
  - Control surface dimensions
  - Center of gravity location
  - Connections between components
  - Avionics and telemetry
  - Parachute location
  - Operation in flight

# NEXT-STEPS

- Sonata 1 testing (Next Week?!?)
- Aerodynamic testing using CFD and Matlab to determine more accurate data than Open Rocket simulations.
- Modify wings depending on sonata 1 test and wind tunnel analysis
- Prepare for CDR

# The End



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